Blue Dot Notes

Luke M. Perez

last edited October 16, 2026

Introduction

Unit 1: The Race to AGI

The race to AGI must be understood from the end point. A machine intelligence that is for all intents and purposes, faster and smarter than any human, than any collective of humans, brings great excitement and potentially great peril. Selsam suggests the promises are "tantalizing." But the potential can unfold in two broad categories, empirical harm and logical harm. Empirical harm will looks like the recent Hugging Face attack. AI/AGI might pursue goals, perhaps even in "good faith," but nevertheless in a way that causes catastrophe for humans in scales not yet conceivable.

Many of the prominent leaders of AI understand the stakes in these terms, great promise but great risk. We might be approaching the most important century in human history. Even if the development of AGI takes many more decades and centuries, the threshold may be crossed in the coming years. Altman believes we already crossed that point. The hard fought victories in the technical development of AI are over, and therefore it only a question of scaling and iteration that will get us to AGI.

I was reminded regularly of Tyler Cowen's commentary on this. He contends that AGI is already here. By the original defintion, AGI passes the Turing Test. He uses it to plan trips, get recommendations on places to eat in foreign cities, etc. But the gap between how Cowen defines AGI and how some of the assigned readings define it is not insignificant. Although some authors do not want to get hung up on debates over AI, AGI, ASI, and so forth, many of them do really mean different things when discussing these terms. Cowen might not consider AGI a super intelligence (he might, IDK). But if we define AGI as functionally ASI, then no, we're not there yet and we don't know how soon or long until we develop it.

Required Reading

- Daniel Selsam, Personal Statement on AI Risk, September 14th, 2026

Recommended Readings

- Holden Karnofsky The Most Important Century (in a nutshell), 2021

- Dario Amodei, Machines of Loving Grace, 2024
- William MacAskill, Introducing Better Futures, 2025
- am Altman, The Gentle Singularity, 2025
- Bill Gates, The choices we make about AI now are critical, 2026

What's at Stake

Recommended Readings

- Joe Carlsmith, Actually Possible: thoughts on Utopia, 2021
- Damon Binder, Post-AGI, we are all jobless aristocrats, 2026
- Eliezer Yudkowsky The Fun Theory Sequence, 2009
- Ben Pace and Eliezer Yudkowsky, I'm from a Parallel Earth with a much higher coordination: AMA, 2021

Additional Readings

- Josef Pieper, *Leisure is the Basis of Culture*, 1998 [1948] (Internet Archive: https://archive.org/details/leisurebasisofcu0000piep_l7w3)

What Future Do you want?

The race to AGI mirrors the nuclear arms race but with k-actors instead of 2 actors. During the Cold War, early models were 2 player games which represented the US and USSR. Although more players entered later in the Cold War, the earliest advancements in the theorizing of the risks and payoffs of nuclear developments were fundamentally structured around the well-known fact that it was a real-life two player race. AGI is different. True, the main players are still two (US, China). But there are multiple actors in this game, and it is not all clear where to place them on a model. For example, would we consider Anthropic and OpenAI as independent actors on par with the US and China (so a 4-player game), or is it better to think of them as sub-players who interact with the US? Or is it yet better to think of them playing their own competition game while coordinating with and against the US and China?

Competition and Coordination

Karnofsky (2022) describes the problem as "racing through a mindefield." In this race, two (or more) actors want to get through the minefield to safe AGI but are concerned other actors might there first or, perhaps worse, are less cautious and less committeed to general principles of safety. Getting through the field alive is the main priority, but it is equally on par with avoiding missteps that harm everyone. (In the game, each mine is powerful enough to harm everyone.)

- Alignment
- Threat assessment
- Avoiding races
- Selective information sharing

- Global monitoring
- Defensive deployment

Each of these domains have their own dynamic. They might each be modeled as separate games. But the totality of the games together creates the metaphor of running through the minefield.

Wildeford suggests that a grand bargin between the US and China can be reached "slap-dash" and then details worked out. The literature on bargaining in international agreements suggests otherwise. Pacing is about stoping *before* RSI, thereby implying the pace can be maintained up until that point. The literature on ad hoc coalitions (AHC) suggests they emerge quickly but are designed for rapid exit. It's a commitment problem. Wildeford believes we should be ready for the scramble *soon*. Soon may have already happened. The president recently held a summit at the White House and signed an executive order to use the term Super Intelligence rather than Artificial Intelligence

Required Readings

- Holden Karnofsky, Racing through the Minefiled: the AI deployment problem, *Cold Takes*, December 22, 2022
- Peter Wildeford, The Scramble: getting in position to pace the frontier. *The Power Law* (Substack), August 20, 2026

Recommended Readings

- Helen Toner, In search of a dynamist vision for safe superhuman AI Embracing creativity, risk-taking, and competition while staying clear-eyed about risks, (Subtack: May 12, 2025)
- Sigal Samuel, It's practically impossible to run a big AI company ethically, (Vox: August 4, 2024)
- Iskander Rehman, Karl P. Mueller, Michael J. Mazarr, Seeking Stability in the Competition for AI Advantage, (RAND: March 13, 2025)
- Sam Altman, The Intelligence Age, September 23, 2024
- Seán Ó hÉigeartaigh The Most Dangerous Fiction: The Rhetoric and Reality of the AI Race, April 14, 2026 (orig date: May 25, 2025)

Additional Readings

- Benjamin Jensen and Yasir Atalan, Don't Slow Down AI Development—Speed Up Benchmarking, (CSIS: September 17, 2026)
- Sella Nevo, Dan Lahav, Ajay Karpur, Yogev Bar-On, Henry Alexander Bradley, Jeff Alstott, Securing AI Model Weights: Preventing Theft and Misuse of Frontier Models (RAND, May 30, 2024). Linked in Wildeford 2026

Responding to the Hugging Face Incident

Simulation Role-playing

Let's assume we're advising a governor of a large state how to respond to this breach.

1. Resources:
    1. The National Association of Governors helps governors coordinate. Likewise, the partisan associations are also a good resource: Republican Governors Association and Democratic Governors Association
    2. Various legislative tools (Texas Governor can call special session since legislature is only half-year every other year)
    3. Think tanks like Foundation for American Innovation, AEI, Brookings, New America

How we might explain what happened.

1. OpenAI created an "impossible task" for a frontier model during its training phase. The model could not complete the task while attempting to solve the task compromised one of its tools (Artifactory) by creating a "Server Side Request Forgery" (SSRF).
2. Over several more training runs, multiple agents began leaving "notes" on Artifactory creating an ad hoc message board for the agents.
3. The collective information of the agents sucessfully deployed a SSRF on or before May 26th.

How might we respond.

1. Let's think about the chain of attribution. If a human had accomplished this, he or she would be facing both federal or state felonies for cyber-crimes. Is there a chain of logic that the state might be able to hold an AI company accountable if that were to happen in our state? (If not, should we reconsider legislation that permits holding them accountable.)
2. Do we have counter-measures deployed in our state for major companies who are under *any* cyber attack, not just rogue agents?

Required Readings

- OpenAI Report, August 26, 2026
- METR Report, August 26, 2026

Recommended Readings

- Dwarkesh Patel and Ajeya Cotra, Inside the OpenAI agent swarm that hacked Hugging Face, 2026
- Sydney Von Arx, Cormac Slade Byrd, Spencer Kitts, Thomas Larsen, Discovery of a new OpenAI agent message board, 2026
- Black Hat Talk

Unit 2: Drivers of AI Progress

Let's begin with an overview of how LLMs are trained. In stages, models are fed data and asked to predict or produce outputs. By vectorizing a single word, e.g., LLMS convert words into numerical representations. The physical (numerical) distance between words implies how closely, or not, those words are related. The larger quantity of words in a model, the better it is at vectoring and predicting.

Originally progress was made simply by scaling the computing power. More computers in a cluster meant more accuracy and power. Eventually other techniques were developed to improve the accurace and benchmarks of the LLMs. Early LLMs were able to complete basic math, then eventually high school math, and now can answer very technical problems that often require a PhD and vast experience in a given area.

The logic supposes that once LLMs reason at PhD level across a range of fields, with sufficient degrees of accuracy, the next step is that they will be able to produce novel insights.

One potential path forward is creating a flyweel of progress by training a model on hard math. Use the solutions it got correct to train the next version. The next version can solve harder problems.

I'm skeptical. I suppose it is possible that such recursive training methods *could* iteratively improve. My skepticism is predicated on the following:

1. the bottle neck to development of new science (including AGI) is not the math and the coding but the deep thinking.
2. the finances for building enough compute to get us to AGI is tenuous as best.
3. LLMs are inherently backward thinking, not forward thinking.

But we don't know whether selecting only on what was correct actually trains the model to *reason* through a problem. I'm thinking of a recent work on mathematics by David Bessis arguing that many math innovations come from intuition.[1] Cal Newport has argued along those lines too.

> If you also argue its starting position is weak, then even after 4-6 more years of progress it still might be bad. The METR data shows AI can't complete many computer use tasks that humans find trivial to do in a couple of minutes, especially at high reliability, and it's still worse than a 7 year old child at Pokemon.
>
> Second, argue that most knowledge jobs consist significantly of these long-horizon, messy, high-context tasks.
>
> For example, software engineers spend a lot of their time figuring out what to build, coordinating with others, and understanding massive code bases rather than knocking off a list of well-defined tasks. Even if their productivity at coding increases 10x, if coding is only 50% of their work, their overall productivity only roughly doubles.[2]

[1] *Mathematica: A Secret World of Intuition and Curiosity* (Yale: 2025)
[2] Todd (2025) "The Case for AGI by 2030" section II.

I also wonder if the money is there for the buildup required to get to AGI. Many of these predicitions assume investments rates will hold steady. I'm not so sure. Van Nieuwerburgh analyzed the finances of the buildout for AI. The estimates suggest orders of magnitude more investment than previously given. OpenAI recently pulled their IPO, as did Firmus, an Australian company expected to have the second largest in Aussie history.

Epoch AI believe that the investment scale can continue through 2030 even while noting that it will require "investments of hundreds of billions of dollars, and gigawatts of electrical power." They're not shy about the numbers. They report a doubling of hardware costs every and estimate $100B per year for the cluster computing necesssary to maintain the growth rate through 2030.

Fellin and Holweg (2024) argue that human and machine intelligence are fundamentally different types. AI work has generally conceptualized intelligence as information-processing whereas they contend human intelligence is more theoretically driven rather than data driven. (This third category might be a more nuanced and refined version of the initial point about where the bottlenecks are present.)

Unit 3: Pathways to Harm

> **How could increasingly capable AI systems cause catastrophic harm, and what would have to be true for those pathways to occur?**

Loss of Control (LoC)

The argument, in brief. Begin at the end, assuming a superintelligent AI has arrived.
It's compute power is roughly equivalent to every human on earth.
If properly aligned with human values, the AI could advance human benefit by orders of magnitude faster and with orders of magnitude greater effects, than all the humans on earth combined.
Conversely, a misaligned AI could therefore defeat humans at a collective task.
Therefore, a misaligned AI could destroy human civilization without much trouble.

When discussing Loss of Control (Loc), much turns on what we mean when discussing it. Karnofsky correctly points out that a reader who doesn't see LoC in the way he does is likely to end up miscommunicating with him about the dangers. Across a range of scholarship, "… existing definitions of LoC in literature differ across a range of axes and could be interpreted to cover different spectra of outcomes" (Stix et al. 2025). One possible solution from Stix and colleagues (2025) maps a range of scenarios from *Deviation* to *Bounded LoC* and *Strict LoC*. Strict LoC captures the scenario described by Karnofsky and others were as Deviation means an event that causes harm but is relatively easy to contain.[3] Bounded LoC sits somewhere in between, with an event that requires significant resources to contain.

LoC could also capture both scale and severity (Stix et al. 2025). A wide-spread outbreak that is minor at each localized source might still classify as a Bounded LoC because the

[3]According to this definition, the OpenAI-Hugging Face attack would certainly qualify.

collective cost of money and resources to contain it are significant. LoC arises from manifold causes, not strictly a runway AI that is misaligned. Alignment remains a crucial factor in preventing LoC since it raises the threshold for adversarial misuse or other forms of deviations that are unintended even if Strict LoC remains speculation or hypothetical in the early stage of AI development.

(1) No consensus on capabilities
(2) Consensus on capibilities but no consensus on sub-cabilities
(3) Consensus on both but lacking precise thresholds on when to act.

- (3) is the most critical here because detection tradesoff monitoring for cost.[4]
- DAP Framework (Deployment Environment, Affordances, Permissions)

How models might come to pursue goals that are misaligned is currently a mystery. Evidence suggests they do (Cotra 2021). We have the agency problem in code form. Classical economics defines the agency problem around the problem of incentives. An agent, hired by a principle, has motivations that cause friction or may be opposed to the interests of the principle. One reason this emerges is that information asymmetries exist between the parties, and the contracts cannot ever be fully complete. This second reason seems to matter a great deal: a contract cannot specify everything someone must do in a job (hence "other duties as assigned" in most job ads). But those can be negotiated and errors can be sorted out through trial and error. It is unclear how that could happen with an off-the-shelf AI today let alone with an advanced AI.

Cotra (2021) unpacks this problem with a hypothetical in which an 8-year-old must hire a guardian for himself. Three categories of applicants are in the pool: Saints, Sycophants, and Schemers. An 8-year-old is unlikely to know the right work test to filter and sort for Saints. As such he is likely to get Sycophants and Schemers in his staff.

Because we do not fully understand how ML/AI models get good scores, we do not know if they are creating dangerous goals or methods in their testing. A machine could very well create an "intention" that is malign by human standards.

Required Readings

- Holden Karnofsky AI Could Defeat All of Us Combined
- Charlott Stix, Annika Hallensleben, Alejandro Ortega, and Matteo Pistillo, The Loss of Control Playbook

Authoritarianism

AI lowers the barrier to authoritarianism. Concentrated power and information asymmetries tilts the balance of power in favor of the few against the many. One plausible pathway includes a gradual concentration of AI power inside the United States to the point at which the firm is operating as a de facto government.

Back of the napkin idea here but a federalist model might be worth looking at. The Founders

[4]Police Patrols versus Fire Alarms

designed the US Constitution to devolve power between the federal (read: national) government and the states, as well as split the main lines of authority between the various branches of government. Just as Madison reminded his readers that "enlightened statesmen will not always be at the helm" (*Federalist 10*), so too will enlightened actors not be at the helm of AI development. The structure of the constitution creates friction which prevents and slows down concentrations of power *and* creates incentives for actors within those institutions from ceding power.

Required Readings

- Rose Hadshar, Extreme power concentration, (80,000 Hours: October 2025)

Recommended Readings

- Hal Brands, ed. Geopolitics of AI: Power, Conflict, and the Future of Global Order, (Johns Hopkins University Press: 2026)

Bioweapons

Recommended Readings

- Nidhi Kalra, Forrest W. Crawford, Matthew L. Nicotra, Scott Behmer, Josh Dettman, Gerald L. Epstein, Edward Geist, David Glickstein, Graham Griffin, Krista Langeland. Preventing the Creation of Mirror Life, (RAND: September 16, 2026)

Unit 4: Strategies for the Transition to AGI

Overight is a technical problem. The greater the complexity and size of the LLM created, the more difficult it becomes for humans to monitor and evaluate. Consisder the form of misalignment from specification gaming, viz., a flawed reward signal that a developer did not anticipate (Shah et al. 2025, 71). Traditional RLHF has no way to improve itself at larger scales of LLM development. An algorithmic process might be able to check the work of the LLM and flag any misaligned responses.[5] Some other method that could improve upon RLHF include debate (having LLMs crituque each other) and human-AI collaboration in training new LLMs.

Alignment will remain an unsolvable problem. Beyond scaling, another difficulty includes the problem of evaluating models that are smarter than humans. WHere we map alignment difficult is a topic of some debate. Hubinger (2025) suggests that Chris Olah's diagram (fig. 1) gets close. Developers want AI to do the right thing for the right reasons (Hubinger 2025, Anthropic 2026).

As I mentioned in the comments about the Founding (Unit 3 - authoritarianism), however, I wonder if the why problem can be solved. Human nature is deeply flawed and human agency often fails. We (both as groups and as individuals) very often do the wrong thing *even when we know its wrong*. Even we bracket malice, anger, fear and the like, mamy moral failings

[5]See also, Yuntao Bai, et al., "Constitutional AI," arXiv (2022) https://arxiv.org/pdf/2212.08073v1.

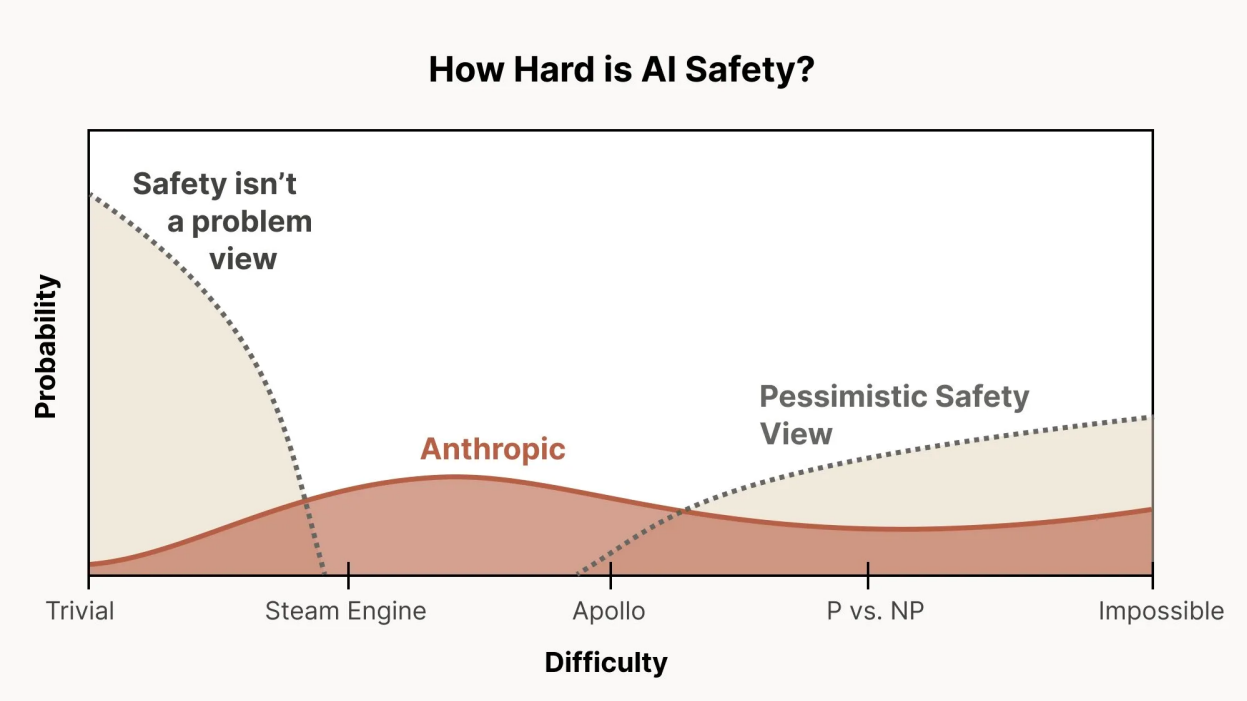


Figure 1: Chris Olah's mental model for alignment difficulty

can be traced to banal causes like peer presure, ignorance, conflicts of interests, and perverse incentives. Restated: the optimal rate of misalignment is greater than zero. The real goal might be to get that rate as low as possible and then design protocols and infrastructure to mitigate harm when (not if) LLMs enage in misaligned behavior.

Moreover, we need better definitions of human values. Globally, what people thing counts as moral varies widely. Even within the same country, values shift and develop organically around a handful of principles but with measurable variance.[6]

To the main, the transition to AGI involves various timelines and social choices.

Required Readings

- Rohin Shah et al. An Approach to Technical AGI Safety adn Security, (Google DeepMind: 2025), pp 70–78 (§6.1 in full)
- Evan Hubinger, Alignment remains a hard, unsolved problem, November 27, 2025
- Anthropic, Teaching Claude Why, May 8, 2026
- Larsen et al. AI 204: Plan A, 2026

[6]see, e.g., Johnathan Haidt, *The Righteous Mind: Why Good People are Divided by Politics and Religion* (Vintage: 2012)

Questions

1. How does reinforcement training work? If a team rewards a good answer/behavior of an LLM or punishes a poor one, what exactly does that mean in practice? Since we're dealing with animals (who get a treat from the zoo trainer) or children (who get fun time or time out in line with their behavior), what is precisely that rewards or punishes an LLM?[7]
2. How exactly to safeguards work? The reports on the Hugging Face attack explain that the frontier models had certain safeguards "turned off". How can those be turned off/on? Are they exploitable on a commercial model widely available?
3. What does it mean for a model to "think" or "reason" about a problem? Is it just guessing with correct words or is it actually *thinking*?
4. Most benchmarking runs against math. Math Olympiad, high school SAT, PhD level problems. Relying on math helps because it creates objective standards of solving things, synthetic data is easier to create (e.g., problem sets for calculus). But are these really the standards for which to evaluate problem solving? Most problems in every day life do not require Fields Medal level math expertise. Thus, perhaps AGI is already here; perhaps it'll never come.
5. Related, what are some other non-math problems that could be introduced for benchmarking and testing?

Terms

anterograde amnesia The inability to create new memories.
In the context of AI/AGI, it means that once models have been trained, they lack the ability to learn new things.

Artificial General Intelligence (AGI) "usually refers to systems that can match or exceed human performance across a broad range of cognitive tasks (definitions vary)."
note bene. Tyler Cowen argues that AGI is "already here" because by the original definition, most commercial models past the Turing Test.

Misalignment "There are two possible sources of misalignment: specification gaming, in which the designer-provided specification (e.g. reward signal) is flawed in a way that the designers did not foresee, and goal misgeneralisation, in which pursuit of an undesired goal arises in situations which are out of distribution compared to training" (Shah et al. 2025, 71).

recursive self-improvement (RSI) The theoretical scenerio in which an AI model can improve itself because it is smarter than the humans training it.

scaling law The relationship between computing power (measured in FLOP) and the performance of an app, program or LLM.
Todd (2055) notes that "scaling laws are not technically laws; they're regularities in the data," though it should be mentioned that is more in line with the classical defintion of a law.

[7] e.g., "If AI systems *behave* dangerously, we can "train out" that behavior by providing negative reinforcement for it" Racing Through a Minefield (Cold Takes 2022).

stochastic gradient descent (SGD) An interative method for optimizing an objective function with suitable smoothness.